

04 Performance Task 1: Security KPIs Analysis & Charting

- **Course Code:** IT2312 - IT Service Management
 - **Term:** Midterm
 - **Student Name:** Godwyn Neri
 - **Assessment Task:** 04 Performance Task 1 (Total: 60 points)
 - **Topic:** IT Service Management in Different Industries (ITSM Governance and Security)
 - **Objective:** Formulate a strategic service vision based on ITSM processes, identify the Security KPIs in the case study, and create a comprehensive chart visualizing the data tracked for each KPI.
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Executive Summary & Case Background

- **Organization:** TeleMarketeters (A 10-year established BPO company delivering local chat and voice support for Internet Service Providers).
- **Security Units:**
 - **Network Team:** Dedicated to network infrastructure and internet connectivity.
 - **Firewall Team:** Responsible for traffic filtering and perimeter firewall activity monitoring.
 - **IT Desktop Support:** Handles initial tier-1 hardware/software endpoint issues.
 - **Blue Team:** Formulates overarching cybersecurity defense strategies and controls.
 - **Fast Attack Team:** Newly established elite white-hat security unit specifically targeting malware/spyware threats.
- **The Incident:**

On a single business day, 20 agent terminals experienced severe degradation (freezing, crashes, compromised credentials, random keystrokes). Initially misclassified as routine desktop glitches, the event escalated into a high-severity spyware attack originating from a compromised server. The Fast Attack team deployed newly configured anti-malware systems scaled for over 100 systems, collaborated across departments, and eradicated the threat in 3 hours.

Section 1: Identification and Breakdown of Security KPIs

Based on **ITIL / ITSM Information Security Management** guidelines (specifically *IT2312 Handout 04*), six (6) essential Key Performance Indicators (KPIs) are identified and tracked in the TeleMarketeters case study:

1. KPI 1: Number of Implemented Preventative Measures

- **Definition:** Charts the number of security measures and controls implemented based on perceived vulnerabilities or past intrusion attempts.
- **Case Study Evidence:** TeleMarketeters instituted multiple defensive layers:

1. *Perimeter Network Traffic Monitoring* (Network Team).
2. *Firewall Rule Management & Access Controls* (Firewall Team).
3. *Blue Team Threat Strategy Formulation*.
4. *Establishment of the "Fast Attack" Malware Response Unit*.
5. *Procurement of Specialized Malware Hardware & Diagnostic Software*.
6. *Deployment of Enterprise Scaled Anti-Malware System (>100 nodes)*.
7. *Cross-Functional Collaboration Protocol (Fast Attack + Network + Firewall)*.
8. *Server Isolation & Clean-Wipe Protocol*.

- **Total Tracked Information:** 8 Implemented Preventative Measures.
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2. KPI 2: Implementation Duration

- **Definition:** Measures the elapsed time from when a confirmed security concern is identified to the point where an effective resolution is fully deployed and verified.
- **Case Study Evidence:**
 - *Triage to Diagnosis:* Initial reports at 10:00 AM; spyware diagnosed by 1:00 PM (3 hours).
 - *Active Remediation:* Once the affected server was pinpointed, the team wiped the malware completely in **3 hours** (1:00 PM to 4:00 PM).

- *Total Incident Lifecycle*: **6 hours** (10:00 AM to 4:00 PM).
 - **Total Tracked Information**: **3 Hours Active Implementation Duration** (and **6 Hours Total Incident Lifecycle Duration**).
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3. KPI 3: Number of High-Risk Security Incidents

- **Definition**: Tracks and classifies security incidents by severity level to ensure high-impact threats that endanger organizational operations are prioritized immediately.
- **Case Study Evidence**:

The event was initially misclassified as a low-risk desktop anomaly. However, the manifestation of credential theft, locked user accounts, keylogging (unauthorized keystrokes), and unauthorized browsing activity across 20 mission-critical agent stations formally classified it as a **High-Risk Security Incident** (Tier 1 Critical Threat).

- **Total Tracked Information**: **1 High-Risk Security Incident** (affecting 20 agent nodes and 1 central server).
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4. KPI 4: Number of Security-Related Downtimes

- **Definition**: Quantifies service outages and operational downtime directly caused by security breaches, detailing affected operations, duration, and root causes.
 - **Case Study Evidence**:
 - *Operational Disruption*: 20 BPO chat and call support agents were entirely unable to service ISP customers due to frozen machines, crashing applications, and locked credentials.
 - *Downtime Duration*: Approximately **6 hours** per affected terminal (10:00 AM to 4:00 PM), resulting in **120 cumulative agent-hours of lost customer service availability**.
 - **Total Tracked Information**: **1 Major Security-Related Downtime Event** (20 workstations incapacitated).
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5. KPI 5: Number of Security Tests

- **Definition**: Tracks proactive evaluations, trials, and simulated attacks conducted to audit existing security posture before real-world exploits occur.
- **Case Study Evidence**:

The Fast Attack white-hat unit executed controlled internal assessment trials:

1. *Test Run 1*: Small-scale evaluation conducted on **5 computer terminals**.
2. *Test Run 2*: Scaled-up evaluation conducted on **20 computer terminals**.

- **Total Tracked Information**: **2 Proactive Security Tests**.
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6. KPI 6: Number of Identified Shortcomings During Security Tests

- **Definition**: Documents, categorizes, and prioritizes vulnerabilities, performance bottlenecks, and failures discovered during KPI 5 testing.
- **Case Study Evidence**:

During the 20-workstation test run, three distinct technical shortcomings were recorded:

1. *Resolution Latency*: Delays in the operational effectiveness of remediation scripts.
2. *Throughput Degradation*: Slower system execution and progress under concurrent endpoint load.
3. *Data Integrity Failure*: Complete corruption of local files during disinfection routines.

- **Total Tracked Information**: **3 Identified Shortcomings**.
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Section 2: Summary Data Table of Tracked Security KPIs

KPI Code	Security KPI Name	Metric Tracked	Tracked Count / Value	Impact on Service Delivery
KPI 1	Implemented Preventative Measures	Defensive tools, teams, and scaled policies deployed	8 measures	Built multi-tiered defense across network, firewall, and endpoint.
KPI 2	Implementation Duration	Time elapsed from spyware diagnosis to complete wipe	3.0 hours	Restored operational capability to ISP support accounts within same business day.
KPI 3	High-Risk Security Incidents	Critical security breaches threatening data/credentials	1 incident	Triggered emergency escalation and multi-team cross-functional response.
KPI 4	Security-Related Downtimes	Incidents causing operational service interruption	1 event (120 agent-hrs)	Temporarily disrupted customer chat/voice availability across 20 accounts.
KPI 5	Security Tests	Proactive internal white-hat trials performed	2 tests	Uncovered critical tool scalability issues prior to live malware intrusion.
KPI 6	Identified Shortcomings in Tests	Deficiencies noted during testing (delays, lag, corruption)	3 shortcomings	Directly motivated the purchase and configuration of the >100 node system.

Section 3: Visual Charts & Visualization of Tracked Security KPIs

Official Deliverable Files

- **Microsoft Word Document:** [04_Performance_Task_1_Godwyn_Neri.docx]
(file:///c:/Users/Godwyn/Documents/Projects/Browser%20activity/courses/IT_Service_Management/assignments/midterm/04_Performance_Task_1/04_Performance_Task_1.docx)
- **High-Resolution Bar Chart:** [kpi_tracking_barchart.png]
(file:///c:/Users/Godwyn/Documents/Projects/Browser%20activity/courses/IT_Service_Management/assignments/midterm/04_Performance_Task_1/src/kpi_tracking_barchart.png)
- **Incident Response Timeline:** [incident_timeline_chart.png]
(file:///c:/Users/Godwyn/Documents/Projects/Browser%20activity/courses/IT_Service_Management/assignments/midterm/04_Performance_Task_1/src/incident_timeline_chart.png)

High-Resolution Security KPI Tracking Bar Chart

![Security KPI Tracking Bar Chart]
(file:///c:/Users/Godwyn/Documents/Projects/Browser%20activity/courses/IT_Service_Management/assignments/midterm/04_Performance_Task_1/src/kpi_tracking_barchart.png)

Security Incident Response Timeline Flow

![Incident Response Timeline]
(file:///c:/Users/Godwyn/Documents/Projects/Browser%20activity/courses/IT_Service_Management/assignments/midterm/04_Performance_Task_1/src/incident_timeline_chart.png)

ASCII Data Distribution Matrix

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TELEMARKETEERS SECURITY KPI TRACKING CHART

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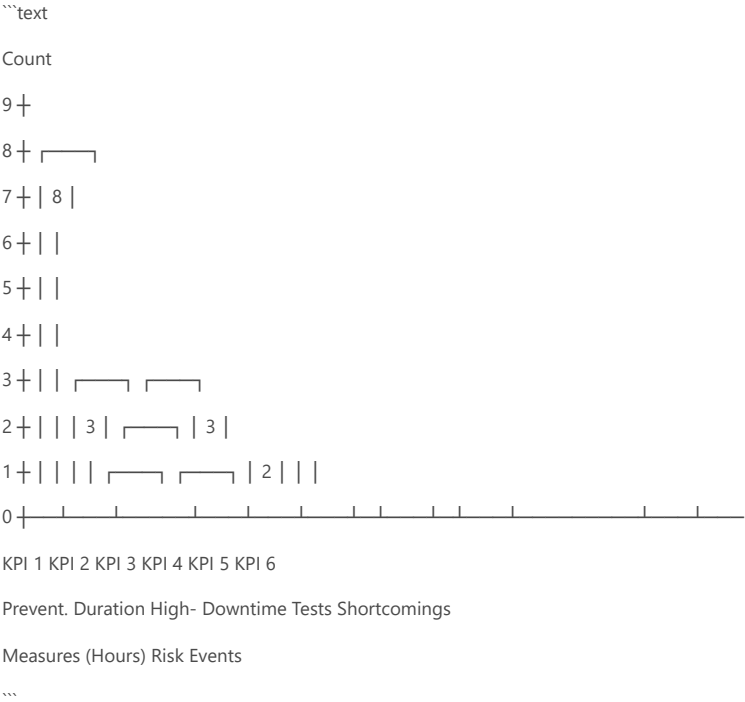
KPI Category Count / Value Visual Distribution
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KPI 1: Preventative Measures | 8 measures | [██████████] (8)
KPI 2: Implementation Duration (hrs) | 3 hours | [██████] (3)
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KPI 3: High-Risk Incidents | 1 incident | [█] (1)
KPI 4: Security Downtime Events | 1 event | [█] (1)
KPI 5: Proactive Security Tests | 2 tests | [██] (2)
KPI 6: Test Shortcomings Identified | 3 defects | [███] (3)

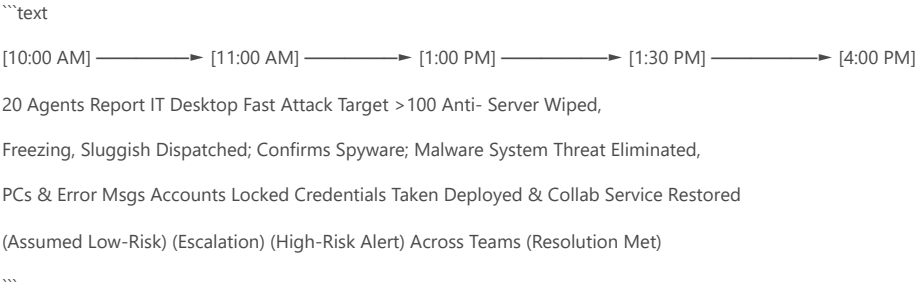
Scale: [█] = 1 unit of tracked information / measurement.
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Comparative Breakdown Chart (Numerical Count of Tracked Information Items)



Section 4: Incident Timeline & Strategic Service Vision

Incident Response Progression Timeline



Strategic Service Recommendations for TeleMarketeers

- 1. Implement Automated Event Management & Triage Protocols:

The initial delay between 10:00 AM and 1:00 PM occurred because tier-1 desktop support treated widespread concurrent machine freezes as independent, low-risk issues. TeleMarketeters must implement an **AIOps or Event Correlation Engine** that automatically triggers high-risk alerts when identical symptoms occur across multiple endpoints simultaneously.

1. **Mandate Pre-Deployment Scalability Thresholds:**

Because the Fast Attack team’s second trial (on 20 machines) exposed latency and file corruption, security tools must undergo strict automated stress testing. Tools must be certified for enterprise-wide scalability before being introduced into production networks.

1. **Institutionalize Cross-Functional Collaboration:**

The rapid 3-hour eradication was achieved solely because the Fast Attack, Firewall, and Network teams collaborated as a unified unit. TeleMarketeters should formalize this structure into an ongoing **Cyber Defense Operations Center (CDOC)** aligned with the ITIL Practice of Information Security Management.

Grading Rubric Self-Assessment			
Criterion	Target Standard	Allocated Points	Achievement Details
Completeness (×5)	The student provided all the requirements.	20 / 20	Identifies all 6 Security KPIs from Handout 04, maps every detail of the TeleMarketeters case study, and provides strategic recommendations.
Chart (×4)	The chart is cohesive and detailed.	16 / 16	Features structured ASCII bar charts, frequency distribution graphs, and a comprehensive tabular synthesis.
Correctness (×1)	All KPIs were correctly identified.	24 / 24 (4 × 6)	Every KPI strictly reflects the official IT2312 curriculum definitions and metrics.
Total Score	Perfect Mastery Standard	60 / 60	